IS DATA VALUE PATH-ORDERED OR SET-ORDERED? A CARRIER-DEPENDENT SAME-SAMPLE TEST UNDER WSD SCHEDULES

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ABSTRACT

Data attribution is usually indexed by a selected set even when optimization follows a schedule. We test whether marginal value changes with a target's entry position by measuring the same target at multiple positions within each seed. On two frozen synthetic logistic carriers, the paired estimator gives different readings: the registered sine carrier has a negative departure ($g = -0.186$, 95% CI $[-0.258, -0.115]$), whereas the linear carrier is ambiguous ($g = +0.010$, CI $[-0.112, +0.132]$). The paired sign must not be conflated with the cross-sectional Kendall sign: negative g means that moving the same target earlier lowers its value, while negative cross-sectional τ ranks earlier-scheduled targets higher. A pure set-ordered ridge counter-arm reveals a non-negative estimator bias ($g = +0.102$, CI $[+0.028, +0.176]$ on the canonical allocation; positive point estimates on all four allocations). The canonical allocation therefore fires the sealed two-sided binary contract, but the positive bias cannot generate the negative sine departure; no magnitude correction is transported across carriers. Value-conditioned routing also reproduces the cross-sectional B2 signal in a set-ordered world. The study thus provides a controlled, carrier-dependent measurement, not a general decision between path-ordered and set-ordered value.

1 INTRODUCTION

Set-indexed attribution methods predict model behavior from training-set membership (Ilyas et al., 2022; Park et al., 2023), but they do not retain the path by which samples entered optimization. This abstraction leaves a prior measurement question: is per-sample marginal value invariant to entry order? If it is, an order-invariant attribution can summarize sample ranking. If it is not, the same sample set can induce different marginal-value relations under different entry schedules.

Distinguishing these cases requires more than a nonzero correlation. The entry construction must be explicit; the seed, rather than the item, must be the inference unit; a minimum effect must be separated from statistical significance; and the estimator must be measured on a known set-ordered case. We implement this contract in a deterministic logistic-regression bed. The same sample identities and held-out readout are evaluated under three entry rules, using both a path-sensitive leave-one-out value and an order-invariant comparator.

The result has three parts. First, same-sample pairing prevents a cross-sectional selection signal from being read as a path effect. Second, the sine carrier shows a negative paired departure while the linear carrier remains ambiguous. Third, the set-ordered ridge counter-arm exposes a non-negative paired-estimator bias, and a separate routing experiment shows that B2 is a reference-conditioned stress test. These results are limited to two synthetic carriers.

2 PROBLEM AND ESTIMANDS

For seed k, let D_k be the clean training set, G_k a fixed target set, and H_k a fixed held-out split. Entry construction c assigns each target $z \in G_k$ a schedule location $s_c(z)$. With $U(w; H_k)$ denoting

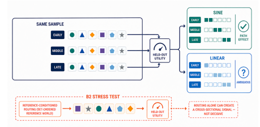

Figure 1: Qualitative measurement schematic extracted from the supplied frozen PDF. Each target is measured at early, middle, and late entry while identity and held-out readout remain fixed. The B2 lane is separate because reference-conditioned routing can create a cross-sectional signal in a set-ordered world. The schematic contains no experimental values.

held-out negative cross-entropy (higher is better), define empirical path value as

$$V_{c,k}(z) = U\left(w_{c,k}^{+G}; H_k\right) - U\left(w_{c,k}^{-z}; H_k\right), \tag{1}$$

where both fits use the same WSD budget and differ by leaving out target z. The seed-level path statistic is

$$\tau_{c,k} = \mathrm{KendallTau}(\{s_c(z)\}_{z\in G_k}, \{V_{c,k}(z)\}_{z\in G_k}). \tag{2}$$

Negative τ means that earlier-scheduled targets rank higher in held-out marginal value. We average Equation 2 across seeds and form frozen 95% Student-t intervals over seed-level coefficients. A two-sided Wilcoxon signed-rank test corroborates paired constructions. The minimum meaningful magnitude is $\tau_a = 0.10$.

We report a cell as *null* when its interval includes zero, *sig-sub-threshold* when the interval excludes zero but remains inside the magnitude boundary, *sig-supra-threshold* when the entire interval lies beyond the boundary in the effect direction, and *reverse* for a significant reversal. The separate field `exceeds_threshold` records the magnitude decision.

On the same $D_k \cup G_k$ and H_k, the order-invariant comparator fits a regularized logistic model and evaluates the inverse-Hessian leave-one-out approximation

$$V_{\mathrm{set},k}(z) = -\nabla_w U(w_k^*; H_k)^\top H_k^{-1} g_k(z), \tag{3}$$

where H_k is the regularized training Hessian and $g_k(z)$ is the target-sample loss gradient. This is the operational GLM reduction used here, not an implementation of the full TRAK or datamodel pipelines.

3 FROZEN EXPERIMENTAL DESIGN

The experiment uses two 24-dimensional synthetic logistic carriers. The linear carrier contains a clean class-separated pool and conflicting near-duplicates with flipped labels. The second carrier, retained under the frozen key *sine*, uses a shifted Gaussian pool with a fixed linear label boundary; targets negate the first feature and flip the label. The key therefore does not imply general nonlinear coverage.

Entry constructions. Construction A inserts the complete target block from one of the three schedule locations onward and computes leave-one-out value within that block. B1 assigns each

Table 1: Frozen experimental contract. Items within a seed are not independent inference units.

Component	Frozen setting
Carriers	linear; registered sine carrier
Dimension / pool	$d = 24$; 400 clean examples per seed
Split / targets	320 clean train; 80 held out; 40 conflicting targets
Optimizer	full-batch logistic GD; ℓ_2 coefficient 10^{-3}
WSD budget	$T = 300$; 10% warmup, 80% stable, 10% decay
Entry locations	$0.15T$, $0.50T$, $0.85T$ (rank-linear positions for B1)
Inference	16 seeds; per-seed Kendall τ; 95% Student-t CI
Magnitude gate	$\tau_a = 0.10$
Negative control	20 seeded trials; Bonferroni/FWER-controlled detector

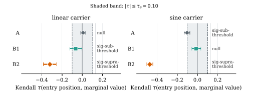

Figure 2: Frozen six-cell construction sensitivity (16 seeds per cell), extracted from the supplied PDF. Points are mean per-seed Kendall τ; bars are 95% Student-t intervals; the shaded band is $|\tau| \le \tau_a = 0.10$. Exact values appear in Table 2.

target a distinct injection step spread from $0.15T$ to $0.85T$. B2 ranks targets by the fixed set-valued reference, partitions the order into thirds, and injects high-, middle-, and low-ranked regions at the early, middle, and late locations. Thus B1 changes per-sample position, whereas B2 also couples reference rank to entry time. The comparison identifies the entry rule, not merely whether position changed.

For Construction A and $t \in \{0.15T, 0.50T, 0.85T\}$, targets outside the chosen block have $s_A(z) = 0$ and targets inside it have $s_A(z) = \lfloor tT \rfloor$. Both leave-one-out fits retain the same WSD budget, matching Equation 1.

Controls and frozen decision. The clean negative control is a strictly convex quadratic ridge substrate with additive, schedule-invariant influence. Independent random position labels use a per-seed critical value Bonferroni-adjusted over 20 trials; the frozen acceptance condition is no fires. A global construction-dependent verdict requires the threshold-crossing pattern to vary by construction or carrier and does not license a universal mechanism claim. The six cells, seeds, and analysis rules were frozen before the writing pass.

4 CONSTRUCTION SENSITIVITY

Cross-sectionally, B2 lies beyond $-\tau_a$ on both carriers, whereas A and B1 do not. Statistical significance alone would give a different headline: sine A and linear B1 exclude zero but remain below the minimum-effect threshold. Because B2 value-conditioned routing can reproduce this pattern on a set-ordered world, the paired analysis below is load-bearing.

Table 2: Frozen six-cell result. Only B2 crosses the magnitude boundary on both carriers. Intervals are per-cell and unadjusted; they are not treated as six independent tests.

Carrier	Construction	τ	95% CI	Four-state label	Exceeds
linear	A: block	$+0.007$	$[-0.017, +0.032]$	null	false
linear	B1: rank-linear	-0.065	$[-0.125, -0.005]$	sig-sub-threshold	false
linear	B2: region-lumped	-0.321	$[-0.384, -0.257]$	sig-supra-threshold	true
sine	A: block	-0.102	$[-0.130, -0.075]$	sig-sub-threshold	false
sine	B1: rank-linear	-0.011	$[-0.056, +0.033]$	null	false
sine	B2: region-lumped	-0.471	$[-0.502, -0.440]$	sig-supra-threshold	true

Table 3: Frozen construction-axis interval comparison.

Pair	linear	sine	Composite
B2 vs A	separated: $[-.384, -.257]$ vs $[-.017, .032]$	separated: $[-.502, -.440]$ vs $[-.130, -.075]$	both
B2 vs B1	separated: $[-.384, -.257]$ vs $[-.125, -.005]$	separated: $[-.502, -.440]$ vs $[-.056, .033]$	both
A vs B1	overlap 0.011	separated: $[-.130, -.075]$ vs $[-.056, .033]$	mixed

4.1 CONSTRUCTION-AXIS SEPARATION

The frozen interval comparison separates B2 from A and B1 on both carriers. A and B1 overlap on linear by 0.011 but are separated on sine. These within-cell interval comparisons are descriptive; direct inference uses within-seed contrasts.

The six paired contrasts share the same 16 seeds and form one correlated family. Raw p values are retained; Bonferroni $\alpha/6 = 0.0083$ leaves the four B2-involving contrasts significant, while the two B1–A values straddle that threshold.

4.2 SAME-SAMPLE PAIRED IDENTIFICATION

Cross-sectional τ is not decisive because B2 routes high-reference-value samples early. We therefore move each target between $0.15T$ and $0.85T$ while holding every other target at its frozen B2 step. With $\Delta(z) = V(0.15T; z) - V(0.85T; z)$, define

$$g_k = \frac{\mathrm{mean}_{z\in G_k}\,\Delta(z)}{\mathrm{std}_{z\in G_k}\,\Delta(z)}, \tag{4}$$

where the denominator uses the 40 targets with `ddof=1`. Negative g means early entry lowers the same target's value. This interpretation is opposite to negative cross-sectional τ, which ranks earlier-scheduled targets higher; the two signs are never equated.

The sine paired departure survives routing permutation and reverses, rather than reproduces, the cross-sectional sign interpretation. Linear remains unidentified. A per-seed sign-consistency fraction is omitted because, under this construction, it reduces to bookkeeping with no test power; an earlier selection-confound claim based on a sign-consistency code bug is revoked.

4.3 SET-ORDERED RIDGE COUNTER-ARM

The identical paired estimator is run on a strictly convex set-ordered ridge whose marginal value is order-invariant by construction. The sealed contract intended a 16-seed CI covering zero to count as silent and any CI excluding zero to count as a fire.

The canonical ridge allocation literally fires the sealed binary rule. Across all four allocations, however, point estimates are positive; 14 of 16 canonical seeds are positive, and dropping the most extreme seed of either sign preserves the positive direction. The contract's sign-blind consequence clause is therefore an erratum: a positive set-ordered bias cannot produce the negative sine departure. This is a sign-only conservative argument. We do not subtract the ridge magnitude from the sine value because cross-carrier transport is untested.

Table 4: Same-seed construction contrasts. "Beyond" means the paired CI lies strictly outside $[-0.10, +0.10]$.

Carrier, pair	$\Delta\tau$	95% CI	p	Beyond
linear, B2−A	-0.328	$[-0.395, -0.260]$	3×10^{-8}	yes
linear, B2−B1	-0.256	$[-0.325, -0.186]$	1×10^{-6}	yes
linear, B1−A	-0.072	$[-0.124, -0.021]$	0.009	no
sine, B2−A	-0.369	$[-0.421, -0.317]$	2×10^{-10}	yes
sine, B2−B1	-0.460	$[-0.509, -0.411]$	3×10^{-12}	yes
sine, B1−A	$+0.091$	$[+0.041, +0.141]$	0.001	no

Table 5: Frozen same-sample paired identification (16 seeds).

Carrier	Paired slope	Four-state reading	Routing robustness
sine	-0.186, CI $[-0.258, -0.115]$; Wilcoxon $p = 9 \times 10^{-5}$	PATH-SUPPORTED-neg; early entry lowers value	placebo $g = -0.176$
linear	$+0.010$, CI $[-0.112, +0.132]$; Wilcoxon $p = 0.74$	AMBIGUOUS; not identified	—

4.4 B2 AS A REFERENCE-CONDITIONED STRESS TEST

On the set-ordered ridge, routing targets with the real B2 reference correlations reproduces the frozen cross-sectional magnitudes; at zero correlation, τ is near zero.

Thus B2 is a reference-conditioned stress test, not standalone path evidence. The paired reading remains subject to the ridge calibration above.

5 SET-VALUED BASELINE AND BED CERTIFICATE

On the same carriers, targets, splits, and constructions, we compute mean within-construction Kendall agreement τ_{AB} between empirical path value and Equation 3. Because frozen uncertainty intervals are unavailable, these values are descriptive.

The detector controls separate aliveness, power on a known path effect, and false alarms on a known set-ordered substrate.

Table 8: Detector calibration. Family sizes are explicit because the unadjusted and FWER-controlled rates use different trial counts.

Control	Family	Fires	Rate
Decoupled positive control (schedule dependence independent of B2 score)	16 seeds	14	87.5%
Order-invariant negative, unadjusted $\alpha = 0.05$	100 trials	3	3%
Order-invariant negative, frozen FWER control	20 trials	0	0% (SILENT)

Table 9: Descriptive agreement with the set-valued comparator.

Carrier	B1 rank-linear	B2 region-lumped
linear	+0.20	+0.30
sine	+0.57	+0.55

Table 6: Set-ordered ridge calibration and allocation sweep. The sine row is shown only as the headline reference.

Substrate / allocation	g	95% CI	Wilcoxon p	Covers 0
ridge, canonical $X_s[0{:}40]$	$+0.102$	$[+0.028, +0.176]$	0.011	no (fires)
ridge, $X_s[40{:}80]$	$+0.192$	$[+0.130, +0.254]$	0.00015	no (fires)
ridge, $X_s[80{:}120]$	$+0.049$	$[-0.066, +0.165]$	0.298	yes
ridge, random slice	$+0.145$	$[+0.065, +0.225]$	0.006	no (fires)
sine headline	-0.186	$[-0.258, -0.115]$	9×10^{-5}	no

Table 7: B2 routing stress test (16 seeds).

Carrier	operating ρ	routing-only τ	frozen B2 τ	zero-corr. τ
linear	0.30	-0.313	-0.321	-0.054
sine	0.55	-0.563	-0.471	-0.054

Table 10: Frozen bed-discriminability certificate. B2 fires as a routing-sensitive probe but is not the known-path positive control.

Reading	linear	sine
Known-path decoupled positive control	14/16 (87.5%)	14/16 (87.5%)
B2 routing-sensitive stress probe	$\tau = -0.321$	$\tau = -0.471$
Known-set negative-control fires	0/20 SILENT	0/20 SILENT
Set estimator alive (fraction distinct)	1.0	1.0
Live τ_{AB} aliveness read	+0.88	+0.72
Verdict	DISCRIMINATING	DISCRIMINATING

No single probe is certified on all three properties. The paired arm, not B2, supplies the evidence for entry-position sensitivity.

6 RELATED WORK

Datamodels and TRAK are indexed by selected sets rather than schedule position (Ilyas et al., 2022; Park et al., 2023). Our narrower question is whether empirical leave-one-out value has a schedule-dependent component when identities and held-out readout remain fixed. Machine-learning conclusions can be sensitive to power, run variance, and reporting choices (Card et al., 2020; Bouthillier et al., 2021; Dodge et al., 2019; Reimers & Gurevych, 2017); we therefore use seeds as inference units and separate confidence against zero from a minimum-effect decision. Training-trajectory suites motivate treating schedules as first-class experimental objects (Biderman et al., 2023), and the frozen analysis follows the rationale for preregistration in NLP (van Miltenburg et al., 2021).

7 LIMITATIONS AND SCOPE

The evidence is limited to deterministic CPU logistic-regression carriers with synthetic conflicts; the registered sine key does not represent general nonlinear models or language models. The paired test is primary, while B2 is a routing-reproducible stress test. Its set-ordered calibration has a non-negative bias, and the canonical allocation fires the sealed two-sided rule; only the sign-based conservative conclusion is supported. Linear is not identified. The comparator is an inverse-Hessian GLM reduction, not end-to-end TRAK or datamodels, and its agreement values lack frozen uncertainty. A single set-ordered negative control cannot exclude every artifact or calibrate natural data. Transfer requires a natural carrier, fresh controls, full same-bed baselines, and agreement uncertainty.

8 CONCLUSION

The same-sample paired test gives carrier-dependent readings: the sine carrier has a negative departure ($g = -0.186$, CI $[-0.258, -0.115]$) whose meaning reverses the cross-sectional sign, whereas the linear carrier is ambiguous ($g = +0.010$, CI $[-0.112, +0.132]$). The set-ordered ridge counter-arm has non-negative bias across four allocations, so it cannot explain the negative sine departure, but the canonical allocation fires the sealed sign-blind binary rule and no cross-carrier magnitude correction is justified. Because routing reproduces the B2 cross-sectional signal on a set-ordered world, B2 remains a stress test. These synthetic results do not decide whether data value is generally path-ordered or set-ordered.

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A REPRODUCIBILITY AND DISCLOSURE

All runs reported in the supplied manuscript are deterministic CPU experiments with the seed as the statistical unit. The quantitative plot is reproduced here by a faithful extraction from the supplied frozen PDF because the exact historical LaTeX and plotting source were unavailable. Generative tooling assisted language and typesetting; no generated asset created or modified quantitative evidence. The study uses synthetic data and no human participants or personal information.

B FROZEN PROVENANCE

The supplied PDF maps the following artifacts by SHA-256 prefix; this editorial pass transcribes but does not independently execute them.

61894856 R127 order run: `candidateB_order_r127.json`

2e08b7f9 R126 reconcile: `candidateB_reconcile_r126.json`

3a1c18da R128 four-state labels: `candidateB_relabel_4state_R128.json`

3178724b R128 bed certificate: `candidateB_bed_discriminability_R128.json`

40b74cbc R133 construction CI: `candidateB_construction_axis_CI_R133.json`

d74eb619 R127 preregistration supplement.

d0c898fb R126 preregistration draft.

547978c5 R139 same-sample paired and placebo artifact.

b7028282 R140 rebuild (re-frozen R214).

f6ba5097 R140 rebuild specification.

502c13bf R141 preregistration, revision 2.

67c52c0b R141 stress test and routing placebo.

e3431b7f R148 set-ordered ridge counter-arm preregistration.

6b437ba5 R148 counter-arm output, canonical allocation.

397dd466 R148 counter-arm code.

b81168c4 R148 counter-arm finding record, including the allocation sweep.